

TAMIL NADU STATE BOARD - STANDARD 11 PHYSICS VOLUME 2

CHAPTER 6: GRAVITATION

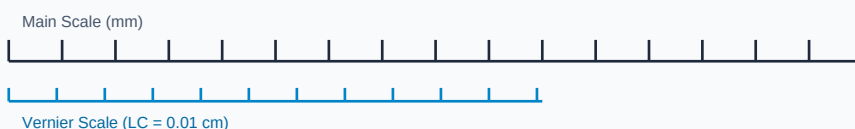
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Physics Volume 2
Chapter Name:	Gravitation

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division

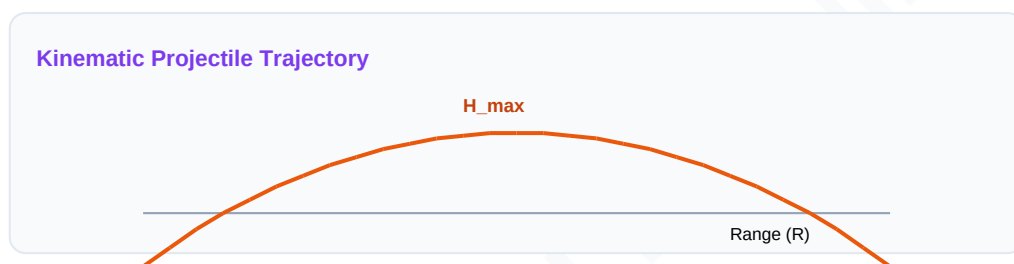


Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Universal Law of Gravitation	The attractive force between two masses is proportional to the product of masses and inversely to distance squared.
Escape Speed	The minimum speed required for a body to escape from the gravitational influence of a planet.
Orbital Velocity	The speed at which a satellite moves around a planet in a stable orbit configuration.

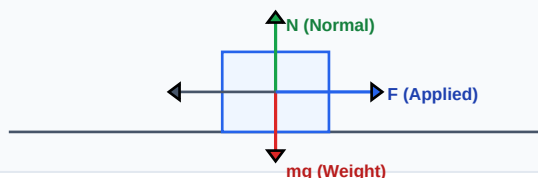
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Force: $F = G * m_1 * m_2 / r^2$ ($G = 6.67 * 10^{-11} \text{ N m}^2/\text{kg}^2$)
- Acceleration: $g' = g * [R^2 / (R + h)^2]$ (altitude variation)
- Escape Speed: $v_e = \sqrt{2 * g * R}$ | Orbital: $v_o = \sqrt{g * R}$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

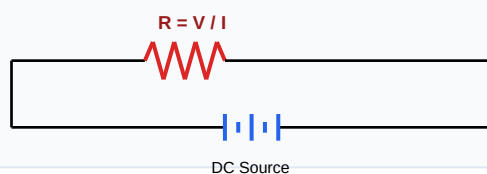
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find escape speed on Earth's surface if radius is 6400km and $g = 9.8 \text{ m/s}^2$.

Step-by-step Solution:

$$v_e = \sqrt{2 * 9.8 * 6.4 * 10^6} = \sqrt{1.2544 * 10^8} = 11200 \text{ m/s} = 11.2 \text{ km/s}.$$

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Derive Kepler's Third Law (Law of Periods) for circular orbits.

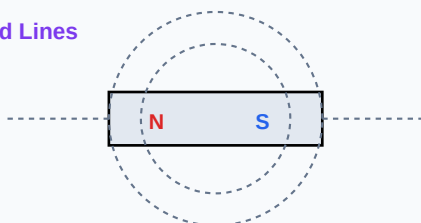
Analytical Solution Strategy:

Centripetal force equals gravitational force: $m \cdot v^2 / r = G \cdot M \cdot m / r^2 \Rightarrow v^2 = G \cdot M / r$.

Since $v = 2 \cdot \pi \cdot r / T$: $(4 \cdot \pi^2 \cdot r^2) / T^2 = G \cdot M / r \Rightarrow T^2 = (4 \cdot \pi^2 / (G \cdot M)) \cdot r^3$.

Hence, T^2 is directly proportional to r^3 .

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Kepler's three laws of planetary motion.

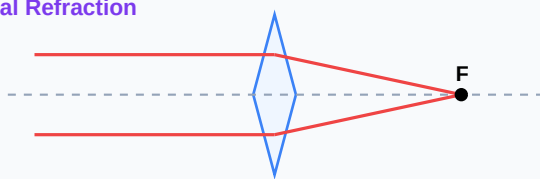
Q2 (2-Mark): How does acceleration due to gravity vary with depth below Earth's surface?

Q3 (2-Mark): What are geostationary satellites? State their time period.

Q4 (5-Mark): Calculate orbital speed of a satellite orbiting close to Earth's surface.

Q5 (5-Mark): Explain weightlessness in space stations.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Satellite Networks

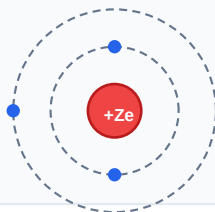
GPS satellites are parked in circular orbits using orbital velocity bounds to sync trackers.

Application Case: Space Exploration

Planetary probes use gravity assist slingshots around planets to speed up without fuel.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

